

Daily Geometry Drill #6 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A Rapid Recognition

1. **Answer:** 40

Method: The angle at the centre is twice the angle at the circumference on the same arc: $2 \times 20^\circ = 40^\circ$.

Investigate further: Halving instead of doubling; the centre angle is the bigger one.

2. **Answer:** 35

Method: $\angle PQR$ and $\angle PSR$ stand on the same chord PR , so they are equal.

Investigate further: Watching the wrong chord; both angles must look at PR .

3. **Answer:** 110

Method: Opposite angles of a cyclic quadrilateral sum to 180° : $\angle C = 180^\circ - 70^\circ = 110^\circ$.

Investigate further: Using $\angle B$ (adjacent, not opposite) instead of $\angle A$.

4. **Answer:** 40

Method: The tangent–chord angle equals the angle in the alternate segment subtended by that chord, so it is 40° .

Investigate further: Quoting the angle's supplement 140° ; the equal angle is on the far side of the chord.

5. **Answer:** 4

Method: Intersecting chords: $PA \cdot PB = PC \cdot PD$, so $2 \cdot 6 = 3 \cdot PD$, $PD = 4$.

Investigate further: Equating $PA = PD$; the products, not the pairs, are equal.

6. **Answer:** 9

Method: Tangent–secant power (Day 5): $PT^2 = PA \cdot PB$, so $9 = 1 \cdot PB$, $PB = 9$.

Investigate further: Forgetting that PAB is P – A – B : PB is the full secant, not the chord $AB = 8$.

7. **Answer:** 90

Method: Thales: a diameter subtends a right angle at the circumference.

Investigate further: Answering 180° (the straight diameter line) rather than the right angle it subtends.

8. **Answer:** 12

Method: Secant–secant power: $PA \cdot PB = PC \cdot PD$, so $4 \cdot 6 = 2 \cdot PD$, $PD = 12$.

Investigate further: Exchanging a near point for a far point; $PB = 6$ is the *full* P – A – B segment.

9. **Answer:** 16

Method: $PT^2 = PA \cdot PB$: $64 = 4 \cdot PB$, $PB = 16$.

Investigate further: Quoting the chord $AB = PB - PA = 12$ instead of the full PB .

10. **Answer:** 30

Method: $\angle BAC$ and $\angle BDC$ stand on the same chord BC , so both are 30° .

Investigate further: Adding the diagonal's split angle; only the chord BC matters.

Section B Manipulation Drills

1. **Answer:** B) 65

Method: Half the central angle: $\frac{1}{2} \cdot 130^\circ = 65^\circ$.

Investigate further: Answering the centre angle itself (130° , option D).

2. **Answer:** A) 80

Method: Opposite angles are supplementary: $180^\circ - 100^\circ = 80^\circ$.

Investigate further: Keeping the angle as-is because the quadrilateral looks symmetric.

3. **Answer:** A) 1

Method: $3 \cdot 5 = 15 \cdot PD$, so $PD = 1$.

Investigate further: Multiplying 3 by 5 and dividing by the nearer point 5, getting a mirror of the given PC .

4. **Answer:** $\frac{25}{2}$

Method: $PT^2 = PA \cdot PB$: $25 = 2 \cdot PB$, $PB = \frac{25}{2}$.

Investigate further: Answering 5 (the tangent half) or dividing 25 by PB instead of PA .

5. **Answer:** B) 55

Method: Alternate segment theorem: the tangent–chord angle equals the angle in the alternate segment.

Investigate further: Taking $90^\circ - 55^\circ = 35^\circ$ (a tangent–radius reflex).

6. **Answer:** A) 6

Method: $PT = \sqrt{PA \cdot PB} = \sqrt{2 \cdot 18} = \sqrt{36} = 6$.

Investigate further: Adding segments into $\frac{PA+PB}{2} = 10$ rather than taking the geometric mean.

7. **Answer:** C) 36

Method: Opposite angles sum to 180° : $2x + 3x = 5x = 180^\circ$, $x = 36$.

Investigate further: Setting $2x = 3x$ or forgetting the opposite pairing.

8. **Answer:** 35

Method: $\angle ACB = 90^\circ$ (diameter), so $\angle BAC = 180^\circ - 90^\circ - 55^\circ = 35^\circ$.

Investigate further: Answering 55° as if the triangle were isosceles.

9. **Answer:** A) 4

Method: Pitot: $AB + CD = BC + DA$, so $DA = 5 + 6 - 7 = 4$.

Investigate further: Pairing adjacent sides instead of opposite ones.

10. **Answer:** 8

Method: $PA \cdot PB = PC \cdot PD$: $6 \cdot 4 = 3 \cdot PD$, $PD = 8$.

Investigate further: Treating PB as a single piece rather than P – A – B ordering.

Section C Substitution & Structure

1. **Answer:** B) 6

Method: Let $CP = 3k$, $PD = 4k$. Intersecting chords: $PA \cdot PB = CP \cdot PD$, i.e. $6 \cdot 8 = 3k \cdot 4k = 12k^2$. So $48 = 12k^2$, $k^2 = 4$, $k = 2$, $CP = 6$ and $PD = 8$. Check: 6 and 8 are consecutive multiples of 3, 4.

Investigate further: Dividing 48 by 7 ($3 + 4$) instead of 12 ($3 \cdot 4$); the power is a product, not a sum.

2. **Answer:** B) 10

Method: Tangent–secant power: $PT^2 = PA \cdot PB$. So $144 = 8 \cdot PB$, $PB = 18$, hence chord $AB = PB - PA = 10$. The diagram is a right triangle OPT with hypotenuse OP .

Investigate further: Quoting $PB = 18$ as the chord, or subtracting the wrong segment ($PT - PA$).

3. **Answer:** C) 6

Method: Ptolemy for cyclic $ABCD$: $AC \cdot BD = AB \cdot CD + BC \cdot AD = 6 \cdot 5 + 4 \cdot 3 = 30 + 12 = 42$. With $AC = 7$, $BD = 42/7 = 6$. Verify $BD < AB + AD$ etc. so cyclic existence holds.

Investigate further: Multiplying adjacent sides ($6 \cdot 4 + 5 \cdot 3$) instead of opposite pairs, or quoting the product 42 as a diagonal.

4. **Answer:** A) $10\sqrt{6}$

Method: $s = \frac{4+5+5+6}{2} = 10$. Brahmagupta: $K = \sqrt{(10-4)(10-5)(10-5)(10-6)} = \sqrt{6 \cdot 5 \cdot 5 \cdot 4} = \sqrt{600} = 10\sqrt{6}$.

Investigate further: Using perimeter 20 for s , or applying Heron to a triangle after an illegal diagonal split.

5. **Answer:** B) 15

Method: Pitot for tangential quadrilaterals: $AB + CD = BC + DA$. So $3x + 5 = 8 + 2x + 3$, giving $3x + 5 = 2x + 11$, $x = 6$, hence $DA = 2x + 3 = 15$. Check opposite sums: $18 + 5 = 23$, $8 + 15 = 23$.

Investigate further: Pairing adjacent sides ($AB + BC = CD + DA$) or solving $3x = 2x$ and dropping constants.

6. **Answer:** A) 6

Method: Secant–secant power from P : $PA \cdot PB = PC \cdot PD$. $PA \cdot PB = 4 \cdot 18 = 72$, so $6 \cdot PD = 72$, $PD = 12$, chord $CD = PD - PC = 6$. Same product 72 is 6×12 .

Investigate further: Using PA as PB (near vs far confusion) or quoting $PD = 12$ as the chord.

7. **Answer:** A) 7

Method: Ptolemy: $AC \cdot BD = AB \cdot CD + BC \cdot DA = 8 \cdot 6 + 6 \cdot 6 = 48 + 36 = 84$. With $AC = 12$, $BD = 84/12 = 7$. Check $7 < 8 + 6$ etc. valid.

Investigate further: Adding adjacent products ($AB \cdot BC + CD \cdot DA$) or dividing the sum by the wrong diagonal.

8. **Answer:** A) 44

Method: $s = \frac{5+5+8+14}{2} = 16$. Brahmagupta: $K = \sqrt{(16-5)(16-5)(16-8)(16-14)} = \sqrt{11 \cdot 11 \cdot 8 \cdot 2} = \sqrt{1936} = 44$.

Investigate further: Using $s = 32$ (full perimeter) or factoring $\sqrt{1936}$ as $22\sqrt{4}$ and miscounting.

Section D Challenge Ramp

1. **Answer:** A) 36

Method: $s = \frac{2+5+10+11}{2} = 14$. $K = \sqrt{(14-2)(14-5)(14-10)(14-11)} = \sqrt{12 \cdot 9 \cdot 4 \cdot 3} = \sqrt{1296} = 36$. Check trapezoid? Also equals Heron split via diagonal 12: two triangles 5–12–13 style.

Investigate further: Forgetting to halve the perimeter, or leaving $\sqrt{1296}$ as a surd.

2. **Answer:** A) 11

Method: Ptolemy: $AC \cdot BD = AB \cdot CD + BC \cdot DA = 7 \cdot 7 + 8 \cdot 9 = 49 + 72 = 121$. With $AC = 11$, $BD = 121/11 = 11$. Both diagonals equal 11; quadrilateral is an isosceles trapezoid in disguise.

Investigate further: Multiplying adjacent sides ($7 \cdot 8 + 7 \cdot 9$) or quoting the product 121 as a diagonal.

3. **Answer:** B) 8

Method: Tangent–secant power: $PT^2 = PA \cdot PB$ with $PB = PA + 10$. So $144 = PA(PA + 10)$, i.e. $PA^2 + 10PA - 144 = 0$, $(PA + 18)(PA - 8) = 0$, $PA = 8$. Check: $8 \cdot 18 = 144 = 12^2$.

Investigate further: Solving $PA^2 + 10PA = 144$ as $PA = 12$ or $PA = 14$ by linear division, or forgetting PB is $PA + 10$ not 10.

4. **Answer:** A) 40

Method: Sides 5, 5, 8, 8 (order $AB = 5, BC = 5, CD = 8, DA = 8$): Pitot check $AB + CD = 5 + 8 = 13 = BC + DA$, so tangential confirms an incircle exists. Semiperimeter $s = 13$. Brahmagupta (cyclic): $K = \sqrt{(13-5)(13-5)(13-8)(13-8)} = \sqrt{8 \cdot 8 \cdot 5 \cdot 5} = 40$. Bicentric: both theorems hold; area also $= r \cdot s$ with $r = 40/13$.

Investigate further: Using Heron on a split triangle, or pairing Pitot incorrectly and claiming tangential fails.

5. **Answer:** A) 19

Method: Ptolemy: $AC \cdot BD = AB \cdot CD + BC \cdot DA = 6 \cdot 12 + 8 \cdot 10 = 72 + 80 = 152$. With $AC = 8$, $BD = 152/8 = 19$. Check $19 < 6 + 8 + 10$ etc. valid diagonal.

Investigate further: Swapping a near-diagonal for a side, or quoting 152 (the sum) as a diagonal.

Top 5 Patterns

- **Product power:** $PA \cdot PB = PC \cdot PD$ and $PT^2 = PA \cdot PB$. Chords crossing, tangent–secant, or two secants from one exterior point all reduce to one product equation. (see A5, A6, A8, A9, B3, B4, B6, B10, C1, C2, C6, D3)
- **Ptolemy adds opposite products.** $AC \cdot BD = AB \cdot CD + BC \cdot AD$ for any cyclic quadrilateral; when one diagonal and all sides are known the other diagonal is one division. (see C3, C7, D2, D5)
- **Brahmagupta is Heron with two extra sides.** $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ with $s = \frac{a+b+c+d}{2}$; the product under the root must be a perfect square for clean integers. (see C4, C8, D1, D4)
- **Pitot pairs opposite sides of a tangential quadrilateral.** $AB + CD = BC + DA$ is necessary and sufficient for an incircle; algebraic sides turn this into a linear equation. (see B9, C5, D4)
- **Cyclic opposites are supplementary and same-segment angles are equal.** $\angle A + \angle C = 180^\circ$ and angles standing on the same chord are equal – the angle-chase that justifies “cyclic” claims. (see A3, A10, B2, B7, C3)

Common Traps

- **Quoting the full secant instead of the chord.** $AB = PB - PA$, not PB itself; tangent–secant always asks for AB you must subtract. (see A6, A9, C2, C6, D3)
- **Multiplying adjacent sides in Ptolemy.** The products are *opposite* pairs $AB \cdot CD$ and $BC \cdot AD$, never $AB \cdot BC$. (see C3, C7, D2, D5)
- **Using the perimeter for s in Brahmagupta.** Semiperimeter is half the perimeter ($s = \frac{P}{2}$); doubling it squares the error. (see C4, C8, D1, D4)
- **Pairing Pitot on adjacent sides.** Pitot is $AB + CD = BC + DA$ (opposites), not $AB + BC = CD + DA$. (see B9, C5, D4)

- **Dividing power products by a sum, not a product.** Intersecting chords: $CP \cdot PD = PA \cdot PB$; with $CP : PD = 3 : 4$ set $CP = 3k$, $PD = 4k$ so $12k^2 = 48$, not $(3 + 4)k$. (see A5, C1, C6, B10)

“A circle trades in equalities; find the chord that both angles stare at, write down the product they share, and the number appears.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths